

Java Programming

SOE104 – Lab 4

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SOE104 Java Programming

Lab 4

Arrays, Static Methods 1, and Testing

Lab Exercise #4.1

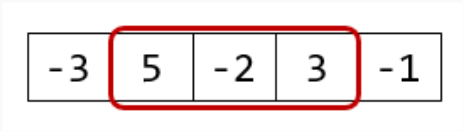
- In this lab, you will learn problem solving using arrays, by creating test cases first

QUIZ
Lab Exercise #4.1 Max Consecutive Sum

- **Complete** the method that takes an integer array and returns an integer:

You are given an array of integers that contains positive, zero, or negative numbers. Your task is to write a static method that finds the **maximum sum** of a sequence of **consecutive elements** in the array and returns this sum.

Write a Java method `int maxConsecutiveSum(int[] arr)` that given an integer array returns the sum of its consecutive elements.



Test cases :

maxConsecutiveSum([-3, 5, -2, 3, -1]) → 6 (the sum of consecutive elements 5 + -2 + 3 = 6 is max)

maxConsecutiveSum([1, -1]) → 1

Lab Exercise #4.1 – Activity 1: Create a Test Class

- Follow the steps in Lecture 4, slides 20 - 26 to **create** a test class
  - by the end, you should have created the following classes:

MaxConsecutiveSum.java × MaxConsecutiveSumTest.java

```java
public class MaxConsecutiveSum {

    public static int maxConsecutiveSum(int[] arr) {


        return 0;
    }

}
```

MaxConsecutiveSum.java MaxConsecutiveSumTest.java ×

```java
import org.junit.Test;
import static org.junit.Assert.*;

public class MaxConsecutiveSumTest {

    @Test
    public void maxConsecutiveSumGivenTestCases() {
        int[] arr1 = {-3, 5, -2, 3, -1};
        int output1 = MaxConsecutiveSum.maxConsecutiveSum(arr1);
        assertEquals( expected: 6, output1);

        int[] arr2 = {1, -1};
        int output2 = MaxConsecutiveSum.maxConsecutiveSum(arr2);
        assertEquals( expected: 1, output2);
    }

}
```

Lab Exercise #4.1 – Activity 2: Create a good set of test cases

- **Add** more test cases to the test class
  - use meaningful test method names

```
4  public class MaxConsecutiveSumTest {
5
6      @Test
7      public void maxConsecutiveSumGivenTestCases() {
8          int[] arr1 = {-3, 5, -2, 3, -1};
9          int output1 = MaxConsecutiveSum.maxConsecutiveSum(arr1);
10         assertEquals( expected: 6, output1);
11
12         int[] arr2 = {1, -1};
13         int output2 = MaxConsecutiveSum.maxConsecutiveSum(arr2);
14         assertEquals( expected: 1, output2);
15     }
16
17 }
```

- Here are some ideas:
  - all positives
  - all negatives
  - all zeros
  - empty array
  - array with one element or two elements
  - array with mix values of large positive and negative numbers
  - array with a single element sum located at the beginning/middle/end
  - ...

Lab Exercise #4.1 – Activity 3: Solve the Problem

- Write your code and continuously refactor it to pass more test cases
  - a simple solution using two nested for-loops is possible
  - *extra challenge:* can you solve the problem with *just* one loop?

- Once you have passed all of your test cases, submit to Learning Mall
  - if your code fails, add more test cases and debug it!

	Test	Expected	Got	
✔	`int[] arr = {-3, 5, -2, 3, -1};` `System.out.println(maxConsecutiveSum(arr));`	6	6	✔
✔	`int[] arr = {1, -1};` `System.out.println(maxConsecutiveSum(arr));`	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 100.00/100.00.

Lab 4 Challenge – Sum of Proper Elements

- In the lab challenge, you will repeat the activities in the lab exercise with a similar problem using an array, method, and testing strategies:

QUIZ
Lab 4 Challenge - Sum of Proper Elements

- **Complete** the method that takes an integer array and returns an integer:

We say that an element of an integer array is **proper** if and only if it is some multiple of (its index + 1).



Write a Java method `int sumOfProper(int[] nums)` that given an integer array returns the sum of all its proper elements.

Test cases :
sumOfProper([1, 2, 6]) → 9 (from 1 + 2 + 6, since all elements are proper)

sumOfProper([10, 25]) → 10 (since only 10 is a proper element)

Lab 4 Challenge – Task 1: Create a Test Class

- Follow the steps in Lecture 4, slides 20 – 26 to **create** a test class
  - by the end, you should have created the following classes:

SumOfProperElements.java | SumOfProperElementsTest.java

```java
public class SumOfProperElements {

    public static int sumOfProperElements(int[] nums) {


        return 0;
    }

}
```

SumOfProperElements.java | SumOfProperElementsTest.java

```java
import org.junit.Test;
import static org.junit.Assert.*;

public class SumOfProperElementsTest {

    @Test
    public void sumOfProperElements() {



    }
}
```

Lab 4 Challenge – Task 2: Create a Set of Test Cases

- **Add** more test cases to the test class
  - use meaningful test method names
  - try to include cases that cover all possible output

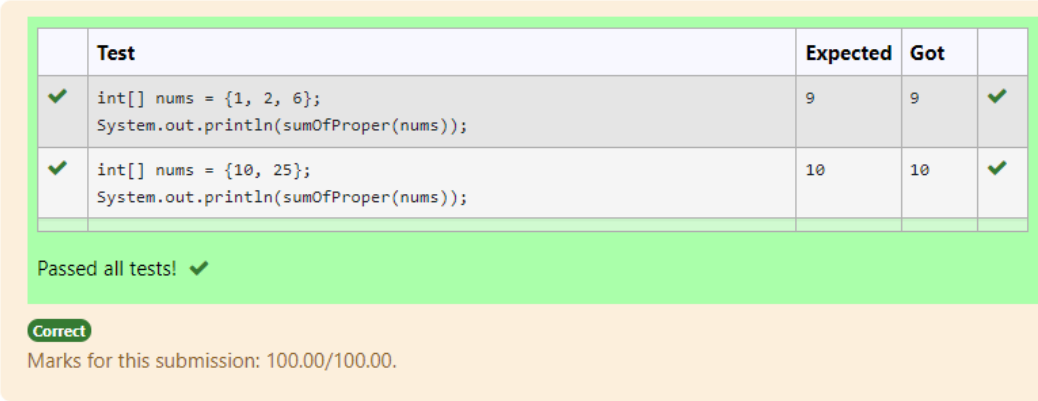
```java
@Test
public void sumOfProperElementsGivenTestCases() {
    int[] arr1 = {1, 2, 6};
    int output1 = MaxConsecutiveSum.maxConsecutiveSum(arr1);
    assertEquals( expected: 9, output1);

    int[] arr2 = {10, 25};
    int output2 = MaxConsecutiveSum.maxConsecutiveSum(arr2);
    assertEquals( expected: 10, output2);
}
```

add your own test cases in new methods below!

Lab 4 Challenge – Task 3: Solve the Problem

- Write your code and continuously refactor it to pass more test cases
- Once you have passed all of your test cases, submit to Learning Mall
  - if your code fails, add more test cases and debug it!
- **Show** to your Lab TA that you have passed the Learning Mall Quiz:



	Test	Expected	Got	
✔	`int[] nums = {1, 2, 6};` `System.out.println(sumOfProper(nums));`	9	9	✔
✔	`int[] nums = {10, 25};` `System.out.println(sumOfProper(nums));`	10	10	✔

Passed all tests! ✔

Correct

Marks for this submission: 100.00/100.00.

- **Briefly explain** to the TA you solution!

Thank you for your effort!

- If you finish early, feel free to use the remaining time to work on other online exercises
  - use this opportunity to practice your new skills in testing your code!

- You may leave the lab once you have complete the Lab 4 Challenge and it has been CHECKED by a TA!